

Multi-class Chest X-ray Disease Classification Using Transfer Learning

Project Report

Dataset: NIH ChestX-ray14 (Custom 7-Class Subset)

Model: DenseNet121 with Transfer Learning

Framework: PyTorch

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1. Introduction

Chest X-ray imaging is one of the most commonly used techniques for diagnosing thoracic diseases. In recent years, deep learning and transfer learning have become widely used in medical image classification, enabling pretrained models to be adapted to various diagnostic tasks.

This project focuses on multi-class chest X-ray disease classification using a pretrained DenseNet121 model. A custom seven-class subset was created from the NIH ChestX-ray14 dataset, including No Finding, Atelectasis, Effusion, Cardiomegaly, Nodule, Mass, and Pneumothorax. The dataset was preprocessed and divided into training, validation, and test sets before model development.

During the project, different training strategies were investigated to improve model performance. The final model was selected based on the experimental results presented later in this report.

This report presents the complete workflow of the project, including dataset preparation, preprocessing, model architecture, training strategy, experimental results, and possible directions for future improvements.

2. Dataset

The dataset used in this project is a custom seven-class subset of the NIH ChestX-ray14 dataset, one of the largest publicly available chest X-ray datasets for thoracic disease analysis. The original dataset contains over 100,000 frontal chest X-ray images collected from more than 30,000 patients and includes annotations for multiple thoracic disease categories.

For this project, seven representative classes were selected:

No Finding

Atelectasis

Effusion

Cardiomegaly

Nodule

Mass

Pneumothorax

A custom subset was created by extracting images belonging to these classes from the original dataset. This subset was then divided into training, validation, and test sets to support model development and evaluation.

Figure 1 presents the seven selected classes, one representative chest X-ray image for each class, and the corresponding number of images in the dataset.

Representative Images from the Selected NIH Chest X-ray Dataset		
Class	Sample Image	Samples
No Finding		10,000
Atelectasis		4,215
Effusion		3,955
Cardiomegaly		1,093
Nodule		2,705
Mass		2,139
Pneumothorax		2,194

Figure 1. Representative chest X-ray images along with the class distribution of the selected seven disease categories.

Finally, the dataset was split into training, validation, and test sets while preserving the class distribution as much as possible.

Table 1 shows the number of images in each dataset split.

Dataset Split	Number of Images	Percentage
Train	21,040	80%
Validation	2,630	10%
Test	2,631	10%
Total	26,301	100%

3. Methodology

The overall workflow of the project is illustrated in Figure 2. The pipeline covers the complete process, from dataset preparation and image preprocessing to model training and performance evaluation.

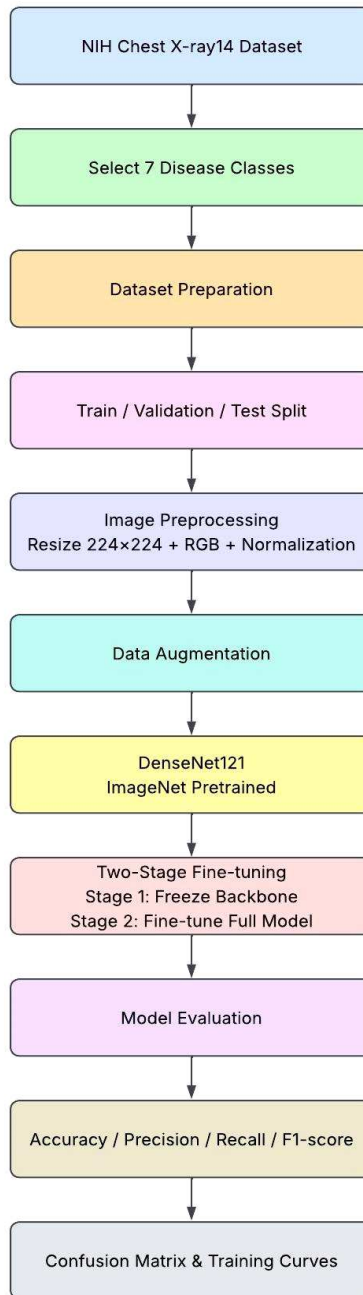


Figure 2. Overall workflow of the proposed chest X-ray disease classification pipeline. The methodology consists of the following stages:

3.1 Dataset Preparation

A custom seven-class subset was created from the original NIH ChestX-ray14 dataset.

The selected images were organized into training, validation, and test directories for model development and evaluation.

3.2 Image Preprocessing

All chest X-ray images were resized to 224×224 pixels before being used for training. Although the original images are grayscale, they were converted to a three-channel RGB representation during data loading to ensure compatibility with the pretrained DenseNet121 model. The images were then normalized using the ImageNet mean and standard deviation.

3.3 Data Augmentation

To improve model generalization, data augmentation was applied only to the training set. The augmentation pipeline included random horizontal flipping and small random rotations. Validation and test images were not augmented.

3.4 Model Architecture

A pretrained DenseNet121 model was used as the backbone network. The original classification layer was replaced with a new fully connected layer containing seven output neurons, corresponding to the selected disease classes.

3.5 Transfer Learning

Transfer learning was adopted by initializing the model with ImageNet pretrained weights. This approach allowed the network to use previously learned visual features while adapting to the chest X-ray classification task.

3.6 Training Strategy

Different training strategies were implemented during model development. The experimental comparison and model selection process are presented in Section 5.

3.7 Model Evaluation

The trained model was evaluated using Accuracy, Precision, Recall, F1-score, and a confusion matrix to assess its classification performance.

4. Experimental Setup

This section summarizes the implementation details and training configuration used in this project.

4.1 Development Environment

Programming Language: Python

Framework: PyTorch

Development Environment: Jupyter Notebook

Hardware: Google Colab (GPU)

4.2 Training Configuration

Table 2 presents the training configuration for the DenseNet121.

Parameter	Value
Input Image Size	224×224
Batch Size	32
Optimizer	Adam
Initial Learning Rate	0.0001
Loss Function	Weighted Cross-Entropy Loss
Evaluation Metrics	Accuracy, Precision, Recall, F1-score

4.3 Training Process

The model was trained in multiple training phases. Different training strategies and optimization techniques were evaluated throughout the development process. Detailed experimental procedures and results are presented in the next section.

5. Experiments

Different training strategies were evaluated during the development of this project to identify a suitable approach for model training.

5.1 Full Fine-tuning

The initial experiments were performed by fine-tuning all layers of the pretrained DenseNet121 model. Although the model was able to learn the training data, the validation performance was not satisfactory, indicating that this approach was not the most suitable for the selected dataset.

5.2 Two-stage Fine-tuning

To improve performance, a two-stage fine-tuning strategy was adopted. The model was trained in three consecutive five-epoch phases, with the best validation performance achieved during the intermediate training stage. Continuing the training beyond this point reduced validation performance, indicating possible overfitting.

5.3 Learning Rate Scheduling

An additional experiment was performed using the ReduceLROnPlateau learning rate scheduler to investigate whether adaptive learning rate adjustment could further improve performance. However, this approach did not lead to a noticeable improvement compared with the previously selected model.

5.4 Final Model Selection

Based on the experimental results, the model obtained during the intermediate stage of the two-stage fine-tuning process was selected as the final model. This model achieved the best overall validation performance and was therefore used for the final evaluation presented in this report.

6. Results

The performance of the final model was evaluated using the test dataset. Multiple evaluation metrics and visualizations were used to assess the classification performance.

6.1 Performance Metrics

The final model was evaluated using the following metrics:

Accuracy

Precision

Recall

F1-score

Table 3 summarizes the overall classification performance.

Metric	Value
Accuracy	50.40%
Precision	54.81%
Recall	50.40%
F1-score	50.87%

6.2 Training Curves

The training and validation accuracy and loss curves were monitored throughout the training process to evaluate the learning behavior of the model.

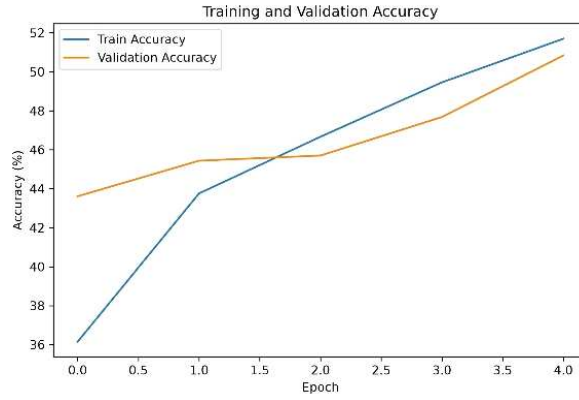


Figure 3. Training and validation accuracy curves over the training epochs.

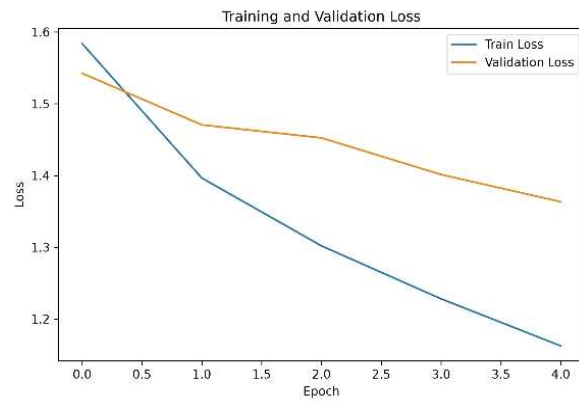


Figure 4. Training and validation loss curves over the training epochs.

6.3 Confusion Matrix

The confusion matrix provides a detailed view of the model's predictions across all seven classes and helps identify correctly classified samples as well as common misclassification patterns.

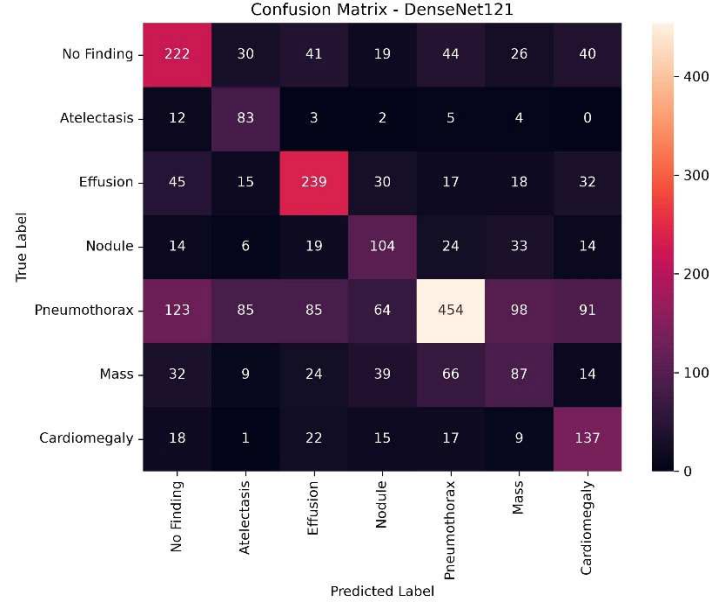


Figure 5. Confusion matrix of the final DenseNet121 model on the test set.

6.4 Classification Report

A detailed classification report, including Precision, Recall, and F1-score for each class, was generated to further evaluate the model's performance.

Table 4 provides the classification performance of the model for each disease class on the test set.

Class	Precision (%)	Recall (%)	F1-score (%)
No Finding	48	53	50
Atelectasis	36	76	49
Effusion	55	60	58
Cardiomegaly	38	49	43
Nodule	72	45	56
Mass	32	32	32
Pneumothorax	42	63	50

7. Discussion

The experimental results show that transfer learning with DenseNet121 is an effective approach for multi-class chest X-ray disease classification. The pretrained model was able to learn meaningful features from the selected dataset and achieve satisfactory classification performance.

During model development, different training strategies were evaluated. Full fine-tuning did not produce the desired validation performance. Adopting a two-stage fine-tuning

strategy improved model generalization and resulted in better validation results. Further training beyond the selected stage did not improve validation performance, indicating that the model had reached its effective learning capacity under the current training configuration.

The class imbalance within the dataset remained one of the main challenges throughout the project. To reduce its impact, weighted cross-entropy loss was used during training. Although this improved the learning process, some classes remained more difficult to classify than others.

Overall, the project demonstrates an effective transfer learning workflow for multi-class chest X-ray disease classification.

8. Limitations

This project has several limitations that may have influenced the overall performance of the model.

The selected dataset is imbalanced, with some disease classes containing significantly fewer samples than others.

Only a seven-class subset of the NIH ChestX-ray14 dataset was used instead of the complete dataset.

The project was developed and evaluated using a single deep learning architecture (DenseNet121).

Due to computational constraints, extensive hyperparameter tuning and comparisons with additional models were not performed.

Despite these limitations, the project provides a complete and reproducible workflow for multi-class chest X-ray disease classification using transfer learning.

9. Future Work

- Several improvements can be explored in future work to further enhance the performance of the proposed classification system.
- Evaluate additional pretrained models, such as ResNet50 and EfficientNet, for performance comparison.
- Perform more extensive hyperparameter tuning to optimize the training process.
- Investigate advanced techniques for handling class imbalance.

- Apply explainable AI (XAI) methods, such as Grad-CAM, to visualize the model's decision-making process.
- Extend the project to include additional disease classes or the complete NIH ChestX-ray14 dataset.

10. Conclusion

This project presented a multi-class chest X-ray disease classification system based on transfer learning using a pretrained DenseNet121 model. A custom seven-class subset of the NIH ChestX-ray14 dataset was prepared, preprocessed, and used for model training and evaluation.

Different training strategies were explored throughout the project. Based on the experimental results, the two-stage fine-tuning approach provided better validation performance than full fine-tuning and was selected as the final training strategy. Additional experiments with learning rate scheduling did not produce a significant improvement over the selected model.

Overall, the project demonstrates a complete deep learning workflow for medical image classification, including dataset preparation, preprocessing, transfer learning, model training, and performance evaluation.